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(1) Research Program/Project/Study Title: Development of Patola-Ginger Jam	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	None
Component Leader:	None
Academic Rank:	None

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Roxas City
Province:	Capiz
Region:	VI

(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☐ Basic	☐ Pilot Testing						
☒ Applied	☐ Tech. Promotion/Commercialization						

(7) Sector/Commodity Agriculture/Fruit and Vegetables	(8) Discipline/Field of Specialization Food Development
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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Patola also known as sponge gourd, fruit shows a good source of calcium and iron and an excellent source of phosphorus. The dried fruit is powdered and made into a snuff for those suffering from jaundice – yellowing of the skin and the whites of the eyes caused by an accumulation of bile pigment in the blood. Patola (sponge gourd) is also noted as a blood purifying agent. It nourishes and protects the liver. It is used for lowering sugar levels in the blood (<http://www.foodrecap.patola-benefits>, Retrieved June 7, 2014).

Ginger is a flowering plant also known as Zingiber officinale. Its root (or rhizome) is a popular ingredient in cooking, but it has also been used for thousands of years for medicinal purposes, particularly in Asian, Arabic, and Indian cultures (*8-health-benefits-of-ginger*/<https://ph.iherb.com/blog>).

The general objective was to develop Patola-Ginger Jam. Specifically, it sought to: 1 determine the sensory qualities of patola-ginger jam in terms of appearance, aroma, taste, consistency and general acceptability among three treatments; and 2. Find out if there is a significant difference in the sensory qualities of the product among three treatments. The study used a Completely Randomized design (CRD). It was conducted in three replicates for sensory evaluation by the twenty (20) semi-trained panelists. Score cards with Nine (9) Point Hedonic Scale was used to obtain the data. The mean and Analysis of variance (ANOVA) were used to analyze the data into alpha level set at 0.01.

Findings revealed that the sensory evaluation of patola-ginger jam, Treatment A got the highest mean score in all quality attributes which described was “Liked Extremely”; by the evaluators across all the qualities. Followed by the treatment B and the lowest was the treatment C.

Furthermore, there were no significant differences in the sensory evaluation of the product in terms of four sensory qualities. This implies that the three treatments of patola-ginger jam were found acceptable regardless of its sensory qualities.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Keywords: Nine (9) Point Hedonic Scale, Sensory Qualities, Jam and Acceptability

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Food is basic to human existence. It is necessary for the health and well-being without it life ceases. Proper nutrition leads to good health. Everyone should be conscious of the food they eat. They should be aware of the nutritional needs and ways on how they can satisfy these. The basic knowledge about nutrition leads to be careful in choosing a healthy food to have a healthy lifestyle. Vegetables provide vitamins and minerals while fruits contain natural sugar, fiber and vitamins that the body needs in order to function properly. Fruits and vegetables are needed in the daily meal to have a balanced diet (*Pinoy Entrepreneur Book A, Diwa Learning Incorporated, Retrieved June 7, 2014*).

Patola also known as sponge gourd, the fruit shows a good source of calcium and iron and an excellent source of phosphorus. The dried fruit is powdered and made into a snuff for those suffering from jaundice – yellowing of the skin and the whites of the eyes caused by an accumulation of bile pigment in the blood. Patola (sponge gourd) is also noted as blood purifying agent. It nourishes and protects the liver. It is use for lowering sugar level in the blood (<http://www.foodrecap.patola-benefits>, Retrieved June 7, 2014).

Many of ginger's professed health benefits have been supported by scientific research: 1. *Aids in Digestive Function*. Compounds in ginger are known to stimulate bile and saliva production and to increase mobility through the digestive tract. Those who suffer from indigestion may find relief by consuming ginger. 2. *May Help Prevent Gastric Ulcers*. Gastric ulcers are painful open lesions on the inside lining of the stomach. Use of non-steroidal anti-inflammatory drugs (NSAIDs), like naproxen and ibuprofen, can lead to the development of gastric ulcers. Bacteria called *Helicobacter pylori* (H. pylori) can also make the gastric lining more prone to ulceration. Ginger, however, inhibits the growth of H. pylori and can prevent the occurrence of stomach ulcers caused by NSAIDs. 3. *Reduces Nausea*.

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Ginger has been shown to be effective for relief and prevention of nausea and vomiting related to pregnancy, chemotherapy, and post-operative conditions. While ginger can help reduce nausea induced by motion sickness, it seems to have no effect on the prevention of vomiting caused by such. Consuming 1 to 1.5 grams of ginger (or 1 to 2 small- to medium-sized pieces of crystallized ginger) can help provide relief for these various types of nausea. 4. *Decreases Pain and Inflammation*. There is evidence that ginger can be beneficial for pain management. By helping to reduce inflammation of joints, ginger can decrease pain associated with arthritic conditions. Daily supplementation of ginger can also improve exercise-induced muscle pain. Women who experience severe pain during menstruation may find some relief with the consumption of ginger as well. 5. *Promotes Sweating*. Ginger is a diaphoretic, meaning that it can stimulate sweating, which allows your body to eliminate toxins. After taking ginger there may be a sense of warming from within, which is why it is popular for soothing symptoms of colds and flu. 6. *Improves Cognition*. With antioxidant and anti-inflammatory properties, ginger can mitigate age-related cognitive decline. There is evidence that ginger can enhance attention and cognitive processing. 7. *May Lower Cholesterol*. There are studies that have demonstrated significant reductions in LDL cholesterol and blood triglyceride levels associated with the consumption of ginger. 8. *May Lower Blood Sugar*. Ginger has been shown to lower blood sugar levels in type 2 diabetic individuals. In the same study, HbA1c (a marker used as an indicator for long-term blood sugar control) also improved. While these are promising results, more research is needed to confirm this benefit (*8-health-benefits-of-ginger/https://ph.iherb.com/blog*).

Nowadays, the changes of lifestyles of many people are moving fast because of the new technologies that provide the necessary needs of people especially in terms of food preparation. Today, preparation of food is more high-tech compared in the past 10 years. Because of the new equipment and machine used by many fast-food establishments, hotels and restaurants all over the world to give absolute satisfaction to their customers. Many chefs continued searching for a new invention and innovation of delicious dishes because of the new standards of food taste. Patola (sponge gourd) and Ginger Jam, the product of this study, when used as filling for sandwich, pastries and biscuits are suited in the modern lifestyle of people because it is very easy to prepare in just a short period of time.

While, patola (sponge gourd) and ginger jam was not yet introduced in the market despite of the fact that it contains high nutrients and abundant in the country. Thus, the conduct of this study will serve as a reference to those who are interested to engage and venture in making jam out of patola (sponge gourd) and ginger for the purpose of family consumption and possible livelihood production.

The general objective was to develop Patola-Ginger Jam. Specifically, it sought to: 1 determine the sensory qualities of patola-ginger jam in terms of appearance, aroma, taste, consistency and general acceptability among three treatments; and 2. Find out if there is a significant difference in the sensory evaluation of the product among three treatments.

The present study will be based on the following prior arts such as:

CN102511704A Fruit jam DALIAN CHUANGDA TECH TRADE MK. The invention relates to a fruit jam, which comprises the following ingredients by weight: 35%-45% of apple juice,; 5%-12% of lemon juice; 1.5% -5% of Chinese yam powder; 3%-5% of Dragon fruit Juice; 2%-4% of yellow peach juice; 3%-5% of poria cocos wolff powder; 2% -4% of donkey-hide gelatin; 0.5% -2% of aloe juice; 1%-3% of food additive, 8% -10% of honey and the balance of pure water. The fruit jam possesses a variety of fruit nutrients, the mouthfeel is moderate sweet and sour, a certain amount of health care components is added, such as Chinese yam powder, donkey-hide gelatin, aloe, poria cocos wolff powder and the like, and frequent eating enables the functions of strengthening the health, discharging toxin and nourishing the skin.

CN105266105A Passion fruit and kiwifruit jam and preparation method thereof. GUILIN BAISHIRUIDONG PASSION FRUIT DEV CO LTD; TANG CHUNYAN. The invention discloses passion fruit and kiwifruit jam and a preparation method thereof and belongs to the technical field of jam. The passion fruit and kiwifruit jam comprises raw materials in parts by weight as follows: 80-100 parts of passion fruit pulp, 30-40 parts of kiwifruit, 10-20 parts of tomatoes, 4-10 parts of hawthorn fruit, 4-10 parts of Andrographis paniculata, 4-10 parts of dahurian angelica roots, 15-25 parts of honey, 2-3 parts of citric acid and 200-400 parts of water. The jam has bright color, moderate sour and sweet, fine, smooth and gentle taste, special fragrance of the passion fruit and high nutrition and health care values; the preparation method is simple; the jam is convenient to eat and low in cost, has broad market

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prospect and is suitable for scale production.

JP2007014314A VEGETABLE BEAN JAM. TAGAMI KAZUYO.PROBLEM TO BE SOLVED: To provide vegetable/fruit bean jam further tasteable while making the best use of taste, flavor and color inherited from a material itself through adding fresh vegetable and fruit to conventional bean jam, increased in palatability and ingestible while easily enjoying nutrient/each kind of vitamin and mineral which are contained in the vegetable and fruit. SOLUTION: The vegetable/fruit bean jam is prepared by mixing vegetable/fruit which are given so as not to lose taste, flavor and color with white bean jam capable of giving accent to the original natural flavor of the material without using processed vegetable/fruit as the raw material so as to make the best use of the original characteristics of vegetable/fruit.

CN108576733A Pineapple jam. LIU ZHENG.The invention provides pineapple jam. The pineapple jam is composed of following raw materials, by weight, 40 to 50 parts of pineapple, 20 to 30 parts of pearl rice, 10 to 16 parts of vegetable chips,10 to 18 parts of honey, 8 to 12 parts of cucumber, 7 to 13 parts of pine nut, 5 to 10 parts of salad dressing, 16 to 20 parts of strawberry, 10 to 20 parts of rock sugar, 6 to 10 parts of licorice root, 7 to 11 parts of dried orange peel, 5 to 7 parts of lemon juice, 4 to 8 parts of pectin, and 30 to 40 parts of water. The pineapple jam is abundant in nutrients; the taste is more sweet, fragrant, and mild; operation is simple and safe; vitamin loss is avoided; abundant nutrients are maintained; the pineapple jam is capable of maintaining human body physiological functions preferably, is suitable to be eaten by people of different ages, and is capable of solving a problem that storage of fruit is difficult.

(11) Materials and Methods or Methodology (This includes the materials, methods,duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in the separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The study used a scorecard as a research instrument. It looked into the four (4) sensory qualities such as appearance, aroma, taste, and consistency as the variables.In scoring the variables of the study, the Nine Point Hedonic Scale using a scorecard was used in evaluating the product.

Statistical Package for Social Sciences (SPSS) was used to generate and process all the data needed in the study. Mean was used to determine the acceptability of the four sensory qualities of the products. The ANOVA was used in order to analyze the differences of the products in four sensory qualities among three treatments.

Three trials were done to come up with the most acceptable product in terms of sensory qualities such as appearance, aroma, taste, consistency and general acceptability. The finished product was evaluated with the semi-trained panelists for sensory evaluation.

The treatments of the study were the following: Treatment A (5grams of Ginger Puree), Treatment B (10 grams of ginger puree), and Treatment C (15 grams of ginger puree) while the rest of the ingredients used in the product formulation were of the same amount such as the patola fruit puree, brown sugar, lemon juice and mineral water. The treatments used in the study varies in the amount of ginger per treatments and the rest of the ingredients were constant.

Product Formulation and Utilization

Procedure in the Preparation of Patola Fruit Puree. Prepare the tools and equipment needed. The young patola fruits were selected and sorted-out. Then, washed it thoroughly in running water and drained. Peeled and sliced the patola fruits and place in the blender and blend until becomes puree. Set aside for later use.

Procedure in the Preparation of Ginger. The ginger was washed thoroughly in running water and drained . Pared the ginger and wash it again. Then, it was sliced into small pieces. Set aside for later use.

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Procedure in the Preparation of Patola-Ginger Jam. Prepare the tools and equipment needed. Measure all the ingredients needed in making the jam. In a blender, combined the patola fruit puree, ginger, brown sugar and blended until the mixture becomes smooth. The puree mixture was placed in the non-stick pan and lemon juice. The puree mixture was boiled on a moderate fire without stirring for 10 minutes. Then, the impurities of the jam was removed. Once the sugar dissolves already stir the mixture continuously to achieve the desired consistency of the jam and to prevent from overcook. The jam mixture was poured on sterilize bottle. Then, it was slightly uncap the bottle and sterilize the jam for 15 minutes. Sealed the bottle cap and let it stay for 3 days before using the jam.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Sensory evaluation of Patola-Ginger Jam

Table 1 shows the preferences of the 30 semi-trained panelists on the sensory evaluation of patola-ginger Jam in terms of appearance, aroma, taste, consistency and general acceptability. The data showed that in terms of appearance, aroma, taste of the product Treatment A, all rated “Like Extremely” and consistency rated as “Liked Very Much”. Followed by Treatment B, and Treatment C got the lowest mean scores among the three treatments.

This result implied that the lower amount of ginger infuse in patola in making jam was preferred by the consumers considering its sensory qualities.

Table 1. Sensory Evaluation of Patola-Ginger Jam.

QUALITY ATTRIBUTES	TREATMENTS					
	A		B		C	
	Mean	QD	Mean	QD	Mean	QD
Appearance	8.40	LE	7.90	LVM	7.50	LVM
Aroma	8.20	LE	7.70	LVM	7.50	LVM
Taste	8.30	LE	7.80	LVM	7.40	LVM
Consistency	8.00	LVM	7.60	LVM	7.70	LVM
Gen.Acceptability	8.23	LE	7.75	LVM	7.53	LVM

Legend:

Scale of Means

8.12 – 9.00

7.23 – 8.11

6.34 – 7.22

Qualitative Description

Liked Extremely - LE

Liked Very Much -LVM

Liked Moderately -LM

Significant Difference in the Sensory Evaluation

Table 2 revealed insignificant differences in the sensory evaluation of patola-ginger jam. The data revealed that there was no significant difference in the appearance in varying treatments as rated by the consumers (F – Value = 2.912, P – value = .062>.01). Therefore, there was no difference exists in the appearance in the three treatments the product, the three treatments were accepted. This implies that regardless of the treatments, the appearance of the products remained the same. As to its aroma of patola-ginger jam did not differed (F-value=2.301, P-value=.109>.01).

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The researchers failed to reject the formulated hypothesis which indicate that there was no significant difference in the product considering three treatments. This means that the treatments were similar in terms of having pleasant aroma. On the other hand, results revealed that there was no significant differences found in taste among the treatments (F-value=2.410, P-value=.099>0.01).

This implies that the taste was acceptable regardless of the variation of the amount of ginger in the jam. The outcome of the test for significant difference among three treatments in terms of consistency failed to reject the null hypothesis (F-value=.760, P-value=.472>.01).

Therefore, the non-existence of a significant of difference in the consistency of the three treatments of the patola-ginger jam was accepted. The evenness in the consistency of the product may due to the similar methods of cooking and ingredients. Furthermore, based on the consumer's acceptability as a whole there were no significant differences among the three treatments of patola-ginger jam with the result shown (F-value=2.249, P-value=.115>.01).

Table 2. Difference in the Consumer's Acceptability of Patola-Ginger Jam.

SOURCES OF VARIANCE	SUM OF SQUARE	df	MEAN SQUARE	F-VALUE	P-VALUE	REMARKS
Appearance	8.133	2	4.067	2.301	.109	ns.
	79.600	57	1.396			
	87.733	59				
Aroma	5.200	2	2.600	2.410	.099	ns.
	64.400	57	1.130			
	69.600	59				
Taste	8.133	2	4.067	.760	.472	ns.
	96.200	57	1.688			
	104.333	59				
Consistency	1.733	2	.867	2.249	.115	ns.
	65.000	57	1.140			
	66.733	59				
Gen Acceptability	5.108	2	2.554			
	64.725	57	1.136			
	69.833					

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(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the findings of the study, the following conclusions are drawn:

Patola Fruit and Ginger can be used as main ingredients in making jam.

Treatment A, B, and C proportion in making jam was acceptable in all quality attributes.

There were no significant differences in the general acceptability of patola-ginger jam considering the four sensory qualities among three treatments.

Recommendations

Based on the findings and conclusions of the study, the following recommendations are forwarded:

Patola and Ginger are recommended as main ingredients in making jam.

All treatments are recommended in making patola-ginger jam, however, individual preferences may vary.

Patola-Ginger Jam may be subjected to microbial and proximate analysis, and its nutritional facts considering potential for product commercialization and technology utilization.

Finally, it is recommended that further studies and applications of patola fruit and ginger may be conducted in order to produce more varied food products aside from jam for consumption and possible livelihood production.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

The output of this study was the production of vegetable based jam which is more healthy and less expensive compared to the existing commercial jam products in the market.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

The product of this study could be one of the sources of extension activities conducted by the University, particularly on the main campus.

(16) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

I would like to thank first the Lord our God who is always the source of my strength and wisdom in the pursuit of this research study. Likewise, I would like to thank all my co-authors, my fellow food technology professors, and evaluators for all the support they have extended throughout the conduct of this study. And also I would like to thank Capiz State University Key Officials, the Department of Industrial Technology, and the Research Office of the campus for the support extended for the success of this research endeavor. To God be the Glory.

(17) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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